P0 Nomenclature Migration Guide

Status: Internal Reference Document
Last Updated: March 15, 2026
Purpose: Explains the evolution of the P0 baseline policy definition for team members interpreting older project artifacts.

Current Standard (v3.0)

Policy	Name	Definition	Code Entry Point
P0	Spatially-Stratified Uniform Allocation	Firehouses sorted by latitude; K evenly-spaced stations selected; units distributed round-robin	`EMSAllocator.baseline_p0()` → `policies.spatially_stratified_allocation()`
P1	Demand-Proportional Allocation	Units allocated proportional to nearby demand	`EMSAllocator.baseline_demand_proportional()` → `policies.demand_proportional_allocation()`
P2	Demand-Weighted MIP Optimization	Mixed-Integer Program minimizing demand-weighted expected response time	`EMSAllocator.solve(model="demand_weighted")` → `models.build_demand_weighted()`

Historical Evolution

Phase 1: Original P0 ("V1")

- **Implementation**: `policies.uniform_allocation()` — round-robin allocation across all 48 firehouses in database (CSV) order
- **Problem**: Firehouse CSV ordering clustered downtown/CBD firehouses first, creating an inadvertent geographic bias
- **Performance**: Mean RT = 8.08 min at K=20, 64.4% 8-minute coverage
- **Period**: Project inception through Phase 8

Phase 2: Spatially-Stratified P0 ("V2")

- **Implementation**: `policies.spatially_stratified_allocation()` — latitude-based spatial stratification
- **Improvement**: Mean RT = 3.17 min at K=20, 99.6% 8-minute coverage (61% improvement)
- **Rationale**: Fair geographic baseline that eliminates CSV-ordering bias
- **Decision Reference**: DEC-011 in `docs/decisions_log.md`
- **Period**: Phase 9 onwards

Phase 3: Nomenclature Standardization ("V3")

- **Change**: Removed all V1/V2 terminology from public-facing documents
- **P0 now always means**: spatially-stratified allocation
- **Original uniform allocation**: Retained in code with `DeprecationWarning`, not referenced in technical report
- **Decision Reference**: DEC-012 in `docs/decisions_log.md`
- **Period**: Phase 11 (current)

How to Interpret Older Documents

If you encounter older project artifacts (notebooks, early draft reports, commit messages), use this mapping:

Old Term	Current Equivalent	Notes
P0 (before Phase 9)	`uniform_allocation()` (deprecated)	Index-based round-robin; **not** the current P0
P0 (V1)	`uniform_allocation()` (deprecated)	Same as above
P0 (V2)	P0 (current)	Spatially-stratified; this is now just "P0"
P0-legacy	`uniform_allocation()` (deprecated)	Another name for the old index-based method
P0-spatial	P0 (current)	Interim name; now just "P0"
P0_baseline	P0 (current)	Code variable name used in some scripts
Production V1	Early experiments using old P0	Results with 8.08 min mean RT
Production V2	Current experiments using current P0	Results with 3.17 min mean RT for P0

Code Compatibility

The deprecated `uniform_allocation()` function remains available but issues a `DeprecationWarning`:

```
# Deprecated — do not use in new code
from ems_readiness.optimization.policies import uniform_allocation
alloc = uniform_allocation(firehouses, K=20)  # DeprecationWarning issued

# Correct — use this instead
from ems_readiness.optimization.policies import spatially_stratified_allocation
alloc = spatially_stratified_allocation(K=20, method='latitude')

# Or via EMSAllocator
allocator = EMSAllocator(...)
result = allocator.baseline_p0(K=20)  # Calls spatially_stratified_allocation internally
```

Performance Comparison (for reference)

Metric	Old P0 (index-based, deprecated)	Current P0 (spatially-stratified)	P2 (optimized)
Mean RT (K=20)	8.08 min	3.17 min	2.57 min
P90 RT (K=20)	19.47 min	5.62 min	3.76 min
8-min Coverage (K=20)	64.4%	99.6%	99.6%
Firehouses Used	20 (arbitrary)	20 (geographically spread)	20 (demand-optimized)

The 61% improvement from the old to current P0 confirms that **geographic placement** is the dominant factor in EMS response time performance.

This document is for internal team reference only. The technical report (`docs/technical_report.md`) presents only the current methodology without historical evolution.